VSP KONDALARAO M

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EDUCATION

Liverpool John Moores University **2010 – 2012**
Masters in Microelectronics *Liverpool, UK*

Pragati Engineering College **2005 – 2009**
B.Tech in Electronics and Communication *Kakinada, India*

COURSEWORK / SKILLS

- HDL Simulation and Synthesis
- Physical Design
- RTL to GDSII flow
- Static Timing Analysis
- CMOS Design

PROJECTS

Tapeout of an Asynchronous CDC FIFO |

- Designed and taped out an 8-bit wide asynchronous FIFO using the sky130 PDK to ensure robust clock domain crossing (CDC), decoupling write and read operations to prevent data loss and metastability between unsynchronized clock domains.
- Optimized silicon area by using a dual-port RAM instead of standard SRAM due to tapeout constraints.

Design of an Application-Specific RISC-V SoC for Clap Switch Control |

- Built a Customizable RISC-V SoC with UART-based instruction bootloader and an on-chip SRAM instruction memory.
- Reduced area from 50K to 15K standard cells by deriving and implementing only the minimal instruction subset required by the application.

TCL Script for Automated Generation of Pre-Layout STA Quality of Results |

- Developed a TCL-based script to parse IO/clock specs from .csv file and auto-generate SDC constraints; Performed RTL synthesis and pre-layout STA.
- **open-source Tools:** OpenTimer(STA), **Yosys** (Synthesis)

Characterization and Integration of a Custom Inverter Cell into the PicoRV32a Core |

- Characterized a custom inverter cell by extracting rise time, fall time, propagation delay, and noise margins using ngspice and Sky130 PDK..
- Integrated the characterized inverter into a custom standard cell library and used it in the full RTL-to-GDS flow of PicoRV32a using Yosys, OpenLane, and Magic.Achieved timing closure using OpenSTA.

TECHNICAL SKILLS

Languages: Verilog, System Verilog, Python, TCL
Tools: Synopsys DC, ICC2, Primetime; OpenLane, Yosys, Magic
Technologies/Frameworks: Linux, GitHub,Git

EXTRACURRICULAR

Circuit Design and Simulation Hackathon by NIT Jamshedpur **Oct 2024 – Nov 2024**

- Designed and simulated Phase-Locked Loop (PLL) blocks using the Sky130 PDK.
- Secured 12th position out of 400 applications. Tools used: xschem, ngspice.

CERTIFICATIONS

- Attended a 5-day workshop on RTL2GDSII flow using Synopsys Tools.
- Semiconductor Fabrication 101 - Purdue University
- VSD Physical Design flow - Udemy.
- VLSI design course - CDAC.
- RISC-V based MYTH processor design - Redwood EDA.